Cyber Resilience of Critical Water Infrastructure: Evaluating a Recent Incident by NIST CSF 2.0

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***Abstract*—Municipal water systems are critical community services whose disruption can directly affect the availability and continuity of essential services for consumers. Cyber security incidents affecting these systems therefore require attention not only to incident containment but also to organizational and technical resilience. This paper analyzes a recent cyber security incident involving the municipal water system in Muleshoe, Texas, using the National Institute of Standards and Technology (NIST) Cybersecurity Framework (CSF) 2.0. The analysis examines the incident through the six CSF 2.0 functions to characterize cyber security practices, incident response, and resilience considerations within a small municipal network environment. Rather than assuming a complex or highly automated infrastructure, the study focuses on the cyber security and resilience implications supported by available evidence from the incident. The analysis identifies areas where the CSF 2.0 functions can provide a structured approach for assessing preparedness, response, and recovery in small municipal water systems. The case also illustrates how cyber security resilience in relatively small local networks can have implications for the continuity and reliability of consumer-facing critical services.**

***Index Terms*—Cyber resilience, NIST Cybersecurity Framework 2.0, critical infrastructure, municipal water systems, cyber security, incident response.**

I. Introduction and Background

Municipal water systems provide essential services and increasingly rely on networked information and communication technologies for operational and administrative activities [1]. A cyber security incident affecting such systems can therefore extend beyond the information technology environment and affect the continuity and reliability of an essential public service [2], [3]. This risk is relevant to small municipalities, where limited resources and relatively small network environments do not eliminate cybersecurity exposure.

The NIST Cybersecurity Framework (CSF) 2.0 [4] provides a structured approach to cybersecurity risk management through six functions, namely Govern, Identify, Protect, Detect, Respond, and Recover. This study applies these functions to assess the cyber resilience implications of a recent municipal water-system incident in Muleshoe, Texas.

On January 18, 2024, unauthorized access to the Muleshoe water system's control environment resulted in changes to operational settings and caused a water storage tank to overflow [5]. The incident was detected by local personnel, after which the affected system was taken offline and operations were shifted to manual control. Reports indicated that the water disinfection system was not affected and that no threat to the public water supply was reported. Fig. 1 illustrates the reported and inferred relationship between external unauthorized access, the HMI, the affected control environment, and the resulting physical impact, however it is not intended as a complete reconstruction of the attack.

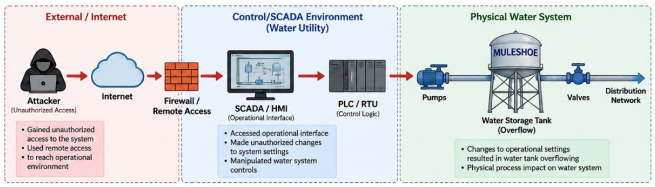


Fig. 1: Illustrated attack pathway of the Muleshoe incident.

This case provides an opportunity to examine how a cyber security compromise in a relatively small municipal network can produce physical consequences while also demonstrating the importance of detection, response, recovery, and operational continuity. The study therefore addresses two questions: (1) How can NIST CSF 2.0 be applied to assess cyber resilience following a cybersecurity incident in a small municipal water system? and (2) What cybersecurity and service-continuity considerations emerge from the incident analysis?

II. Evaluation Using NIST CSF 2.0

The NIST Cybersecurity Framework (CSF) 2.0 organizes cybersecurity risk management into six functions, which provide a high-level structure for organizing cyber security outcomes and are intended to be considered concurrently rather than sequentially. In this paper, publicly available evidence from the Muleshoe water-system incident is considered across the six functions to evaluate their relevance to the reported cyber event, operational disruption, and response.

Because detailed information about the utility's internal architecture, security controls, and cyber security policies is not publicly available, a five-point qualitative relevance scale is used, such as 1 (low), 2 (moderate), 3 (high), 4 (very high), and 5 (critical). The scores represent an incident-specific assessment of the relationship between each CSF function and the available evidence, whereas they are not NIST scores, implementation tiers, maturity levels, or measures of the utility's overall cyber security pictures. As shown in Table I summarizes the incident-based assessment and the primary resilience consideration associated with each CSF Function.

TABLE I: Evaluation of NIST CSF 2.0 functions.

Function	Incident-based evaluation
Govern	Cybersecurity governance and risk-management practices are relevant to the incident, although detailed governance information is not publicly available.
Identify	The incident highlights the importance of identifying remotely accessible systems, critical assets, and their cybersecurity exposure.
Protect	Unauthorized access to the water-system environment highlights the importance of appropriate authentication, credential management, access control, and protection of remotely accessible systems.
Detect	The physical consequence of unauthorized activity demonstrates the importance of timely detection of abnormal or unauthorized system activity.
Respond	Operators isolated the affected system and transitioned operations to manual control, showing the importance of containment and operational response procedures.
Recover	Continued operation and restoration following the incident demonstrate the importance of recovery procedures and maintaining essential service continuity.

In summary, the assessment indicates that Protect and Detect have the strongest direct relationship with the reported incident, while Respond and Recover are central to maintaining operational continuity. On the other hand, Govern and Identify provide the supporting risk-management foundation for improving resilience. The analysis shows how NIST CSF 2.0 can be applied to a relatively small municipal water-system environment without assuming technologies or security controls that cannot be verified from publicly available evidence.

III. Recommendations and Discussion

The Muleshoe incident illustrates that cyber resilience extends beyond preventing unauthorized access. Also, it requires timely detection, effective response, operational continuity, and recovery. Based on the incident and the NIST CSF 2.0 assessment, the following three areas are particularly relevant to small municipal water systems.

Access protection and exposure management: Remotely accessible systems should be identified and regularly reviewed, with appropriate authentication, credential management, access privileges, and externally accessible services.

Detection and incident response: Municipal operators should establish practical mechanisms for identifying abnormal or unauthorized activity and maintain documented procedures for system isolation, communication, and incident response.

Operational continuity and recovery: Critical water operations should have documented manual or alternative operating procedures that allow essential services to continue while affected systems are isolated or restored. Maintaining such alternatives improves resilience by reducing dependence on a compromised network during an incident.

The case also highlights the connection between cyber security and consumer-facing service reliability. Also, it shows how NIST CSF 2.0 can provide a structured basis for examining cyber security resilience, operational continuity, and recovery in small critical-service networks. Although the examined

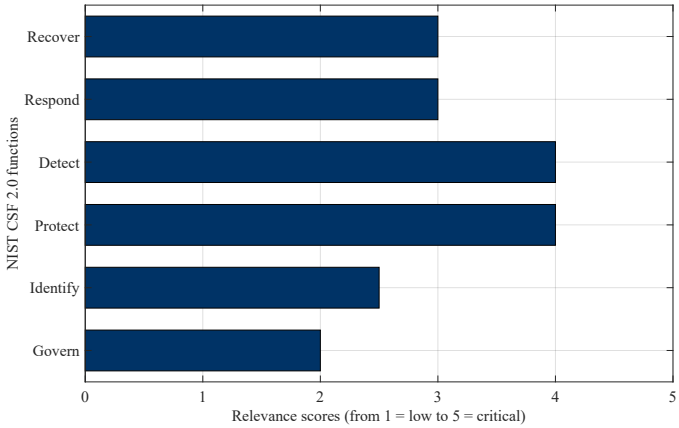


Fig. 2: Incident-based relevance of NIST CSF 2.0 functions.

environment represents a relatively small municipal network, a cyber security compromise can produce consequences beyond information systems when the network supports an essential public service. Cyber resilience therefore includes both protection of networked systems and continuity of the services they support.

However, this work is limited by the availability of public information and does not provide visibility into the utility's complete network architecture, security controls, policies, or incident-response procedures. Accordingly, the CSF relevance scores should not be interpreted as a comprehensive cyber security audit or maturity assessment. Hence, future work can incorporate validated technical and organizational information to develop more detailed CSF 2.0 assessments of small municipal water systems.

Acknowledgements

This work was supported by the US Department of Transportation (USDOT) Tier 1 University Transportation Center (UTC) Transportation Cybersecurity Center for Advanced Research and Education (CYBER-CARE) (Grant No. 69A3552348332); and the National Research Foundation of Korea (NRF) grant funded by the Korea government(MSIT) (No. RS-2024-00463802).

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